

GRB 081125496

AGENTIC AI REPORT BY VIKAS CHAND, KHUSHBOO SHARMA, AND JAGDISH C. JOSHI

ABSTRACT

We present a complete timing and time-resolved spectral analysis of the bright single-pulse burst GRB 081125496 using *Fermi*/GBM data, as the first burst of a 106-burst single-pulse campaign executed with a staged, gate-reviewed pipeline. The windowed duration is $T_{90} = 8.50^{+0.15}_{-0.19}$ s with 5.3σ of emission beyond the analysis window, the pulse is FRED-like (two-sigmoid asymmetry $\varphi = 0.317 \pm 0.026$), and the per-band durations follow $T_{90} \propto E^{-0.14 \pm 0.02}$. The minimum variability timescale depends on the estimator by design: a windowed method finds 33.9 ± 2.9 ms on the pulse rise, while global wavelet methods return 215 ± 17 ms (CWT) and 546 ± 69 ms (Haar) for the same data. The spectral lag between 25–50 and 100–300 keV is $\tau = 0.725^{+0.070}_{-0.094} \pm 0.139_{\text{win}}$ s (soft lagging hard), which for a single broad pulse we interpret as envelope evolution: $\tau/T_{90} \approx 0.08$. We then fit 24 spectral models in nine Bayesian-block time bins and in the time-integrated spectrum, selecting by the Akaike Information Criterion. A cutoff power law suffices in seven of nine resolved bins; a high-energy tail is demanded only on the pulse rise and in the late tail, where the Band function wins with $\beta \simeq -2.1$. The time-integrated winner, a smoothly broken power law plus power law, wins no resolved bin: the integrated “extra component” is spectral evolution in disguise. Across ~ 90 blackbody-added fits we find zero photospheric evidence; every low- kT solution is an edge-constrained artifact at the fitted band’s lower boundary. We discuss the implications of estimator- and window-dependence for population lag–variability correlations. All figures and numbers in this paper passed independent verification gates; the analysis provenance is machine-auditable.

Keywords: Gamma-ray bursts (629) — Time domain astronomy (2109)

1. INTRODUCTION

Single-pulse gamma-ray bursts offer the cleanest laboratory for prompt-emission physics: one emission episode, no pulse overlap, and a light curve whose spectral evolution can be followed from rise to decay without deconvolution. They anchor pulse-paradigm descriptions of GRB variability (Norris et al. 1996; Kocevski et al. 2003), spectral-lag physics (Ukwatta et al. 2010, 2012), curvature-effect tests (Shenoy et al. 2013), and searches for spectral components beyond a single continuum (Ravasio et al. 2019; Yu et al. 2019).

GRB 081125496 (*Fermi*/GBM trigger bn081125496; RA = 42°7, Dec = −18°9) is a bright single-pulse burst with a GBM catalog duration of $T_{90} = 9.28 \pm 0.61$ s and a 10–1000 keV fluence of 1.85×10^{-5} erg cm^{−2}. It is burst #1 of a 106-burst single-pulse campaign whose sample is selected on light-curve morphology, not brightness. This paper presents the complete per-burst analysis — detector selection through time-resolved spectroscopy and timing — and doubles as the reference implementation of the campaign’s staged pipeline: each section below corresponds to one pipeline step, and each step’s prod-

ucts passed through independent verification gates before entering this manuscript (§9).

2. OBSERVATIONS AND DATA

GRB 081125496 triggered the *Fermi* Gamma-ray Burst Monitor (Meegan et al. 2009) on 2008 November 25. We use time-tagged event (TTE) data from the NaI and BGO detectors. Following the convention of Chand et al. (2020), spectral fits use NaI channels in 8–900 keV with the 30–40 keV iodine K-edge region excluded, and BGO channels in 0.2–38 MeV. Data in every band are retained subject to data-quality cuts only; no detection-significance cut is applied, to avoid biasing the census of spectral shapes.

3. DETECTOR SELECTION

The campaign selects NaI detectors with source angles $\leq 50^\circ$ (with a 50–60° rescue for detectors flagged in the burst catalog) and the BGO on the source side. For GRB 081125496 this yields NaI detectors n_a (reference, smallest angle) and n_b , and BGO b_1 (Figure 1). All subsequent analysis uses these three detectors.

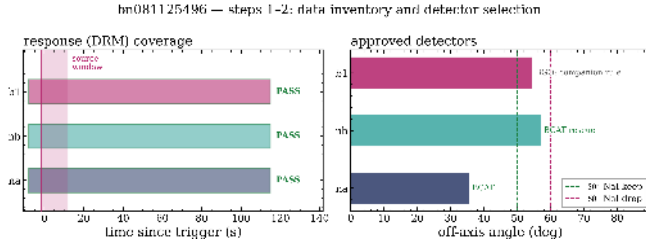


Figure 1. Detector selection for GRB 081125496: source angles for all GBM detectors with the 50° selection rule and the 50 – 60° catalog-rescue band; the selected detectors (n_a , n_b , b_1) and their data coverage.

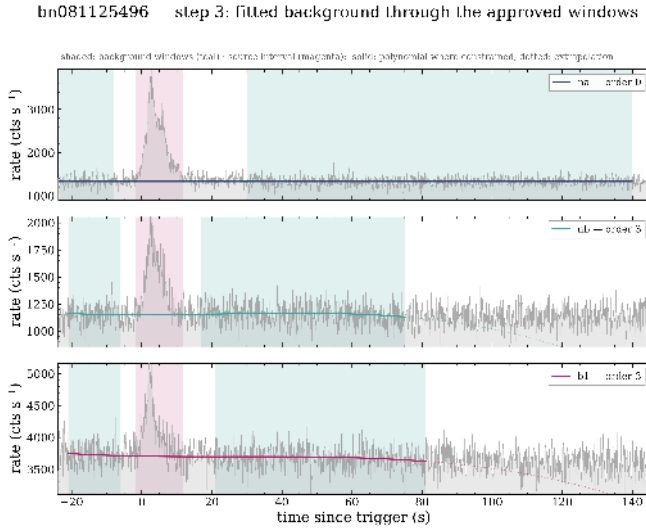


Figure 2. Approved background intervals and polynomial fits for the three selected detectors.

4. BACKGROUND AND SOURCE INTERVALS

Background and source intervals were selected interactively by a human reviewer using a gtbust-style two-panel interface, and are recorded with approval stamps in the campaign catalog. For GRB 081125496 the background is fit on per-detector pre- and post-burst windows (n_a : $(-23.5, -8.0)$ and $(30, 140)$ s; n_b : $(-21.0, -6.0)$ and $(17, 75)$ s; b_1 : $(-21.0, -6.0)$ and $(21, 81)$ s) with polynomial orders selected per detector (0–3), and the approved source window is $(-1.7, 11.9)$ s (Figures 2, 3). The background windows deliberately hug the burst, and the same approved selections drive both the spectroscopy and the timing analysis — a single set of human decisions propagates through every product.

5. TIMING ANALYSIS

All timing measurements run on the approved selections of §4, on the reference NaI; the spectroscopy (§7) uses the same selections.

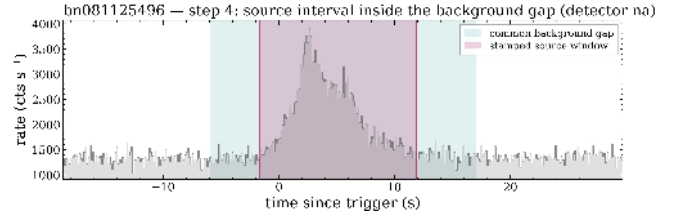


Figure 3. Approved source interval over the background-subtracted light curve.

5.1. Durations

We measure durations inside the approved source window with a first-crossing cumulative estimator; uncertainties come from Poisson realizations of the raw counts propagated through the identical estimator, with the t_5 – t_{95} covariance retained (cf. the quadrature approximation of Qin et al. 2013). We find $T_{90} = 8.50^{+0.15}_{-0.19}$ s spanning $[0.47, 8.97]$ s and $T_{50} = 3.50$ s. Emission continues beyond the window: the interval between the source window and the post-burst background window holds 5.3σ (≈ 790 net counts) of excess, so our T_{90} is a *windowed lower limit*, consistent with the catalog 9.28 ± 0.61 s (Shao et al. 2017) once the excluded tail and the count-space/fluence-space methodological difference are accounted for. The same truncation applies to the spectroscopy: all spectral statements describe the windowed emission.

Per-band durations decrease with energy from 9.4 ± 0.2 s (8–25 keV) to 6.6 ± 0.3 s (100–350 keV), following $T_{90} \propto E^{-0.14 \pm 0.02}$ ($\chi^2/\text{dof} = 0.5$; Figure 4), consistent within 2σ with the population-mean slope -0.20 ± 0.02 of Qin et al. (2013) — which we treat as a consistency check, not a per-burst law (cf. Kouveliotou et al. 1993).

5.2. Pulse Shape

We fit three analytic pulse models to the 8–900 keV net light curve (Figure 5): the FRED form of Norris et al. (1996), the Kocevski et al. (2003) profile, and the two-sigmoid of Gowri et al. (2025). The Kocevski profile wins ($\chi^2_r = 1.04$; Norris 1.08). The Gowri two-sigmoid, fitted under the protocol of that paper ($r_l \leq r_r$, $R^2 = 0.961 \geq 0.7$), gives the shape parameter $\varphi = s_l/s_r = 0.317 \pm 0.026$: a clean FRED-like asymmetry. We note φ is a sigmoid-width ratio and is not directly comparable to rise/decay-time asymmetries.

5.3. Minimum Variability Timescale

We measure the MVT with three estimators that differ at the primitive, and we report all three with labels because they answer different questions (Figure 6):

1. *Windowed (canonical)*: the validated framework of Bala et al. (2026), which scans short segments

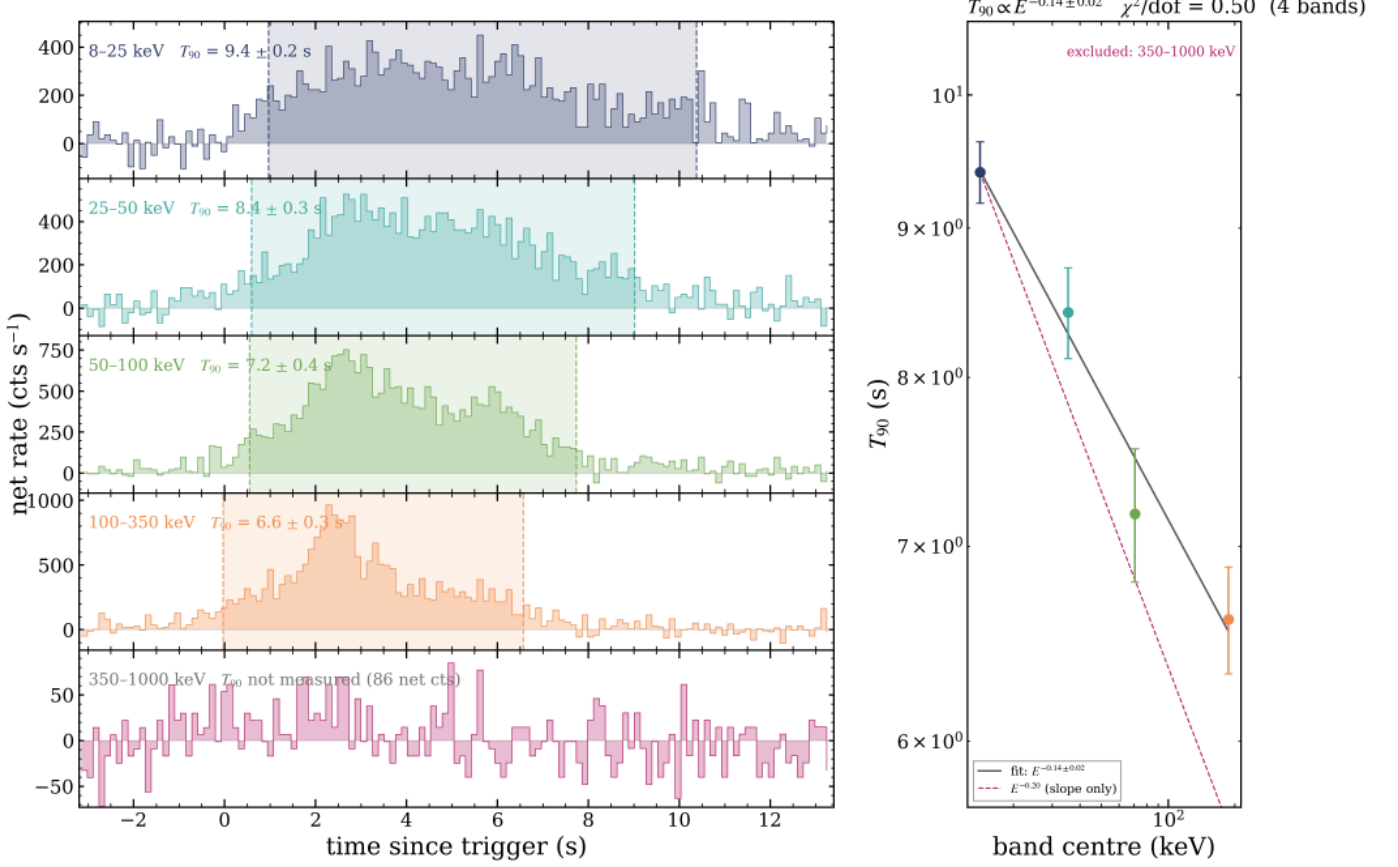
bn081125496 — step 7: energy-resolved T_{90} (detector na); dashed = t_5 , t_{95} 

Figure 4. Per-band light curves with individual T_{90} measurements (shaded t_5 – t_{95} spans; the 350–1000 keV band is excluded for insufficient counts) and the $T_{90}(E)$ relation with the [Qin et al. \(2013\)](#) population slope for reference.

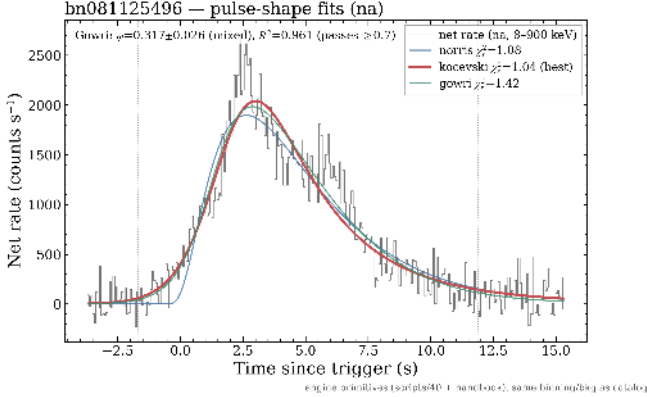


Figure 5. Pulse-shape fits to the net 8–900 keV light curve. The Kocevski profile is the χ^2_r winner; the Gowri two-sigmoid supplies the asymmetry parameter φ .

across the burst, finds 33.9 ± 2.9 ms, localized on the pulse rise (Figure 7). The quoted uncertainty is the per-fit error; an independent re-run shows realization scatter beyond it, and the local detection z -score is 2.5 (weighted-mean significance 6.2)

— we therefore quote this as the burst’s MVT with those caveats attached.

2. *Global CWT*: the continuous-wavelet method of [Vianello et al. \(2018\)](#) applied to the full source window returns 215 ± 17 ms — the timescale at which the *whole window’s* power exceeds its Poisson floor.
3. *Global Haar*: a Haar-wavelet structure minimum (cf. [Golkhou & Butler 2014](#); [MacLachlan et al. 2013](#)) returns 546 ± 69 ms.

The ordering is not a discrepancy but a scope statement: a smooth 8.5 s FRED dominates any global spectrum, while only a windowed scan can isolate the ~ 30 ms structure on the rise. Population studies that mix MVT estimators mix questions; the energy dependence of MVT ([Golkhou et al. 2015](#)) compounds this across instruments.

5.4. Spectral Lag

We measure the 25–50 vs. 100–300 keV lag with the discrete cross-correlation function ([Band 1997](#)), an

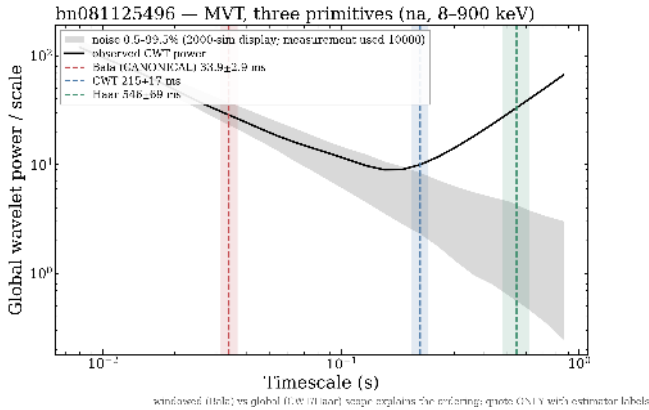


Figure 6. Global continuous-wavelet power spectrum of the GRB 081125496 source window (black) against the 0.5–99.5% envelope of 10^4 Poisson realizations (gray), with all three MVT estimates marked (dashed lines with 1σ bands). The global MVT is the scale at which observed power exits the noise envelope — a characteristic timescale closer to the rise time of the shortest structures than to the shortest observable bin (Golkhou & Butler 2014). The windowed estimate (red) probes rise structure that the whole-window spectrum averages away; the Haar minimum (green) applies a later threshold on the same data.

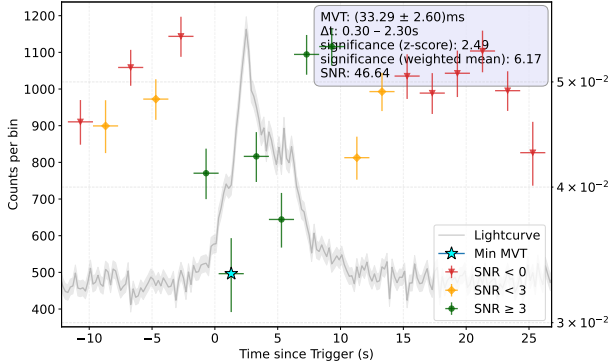


Figure 7. Native diagnostic of the windowed MVT framework (Bala et al. 2026): MVT per 2 s segment over the light curve, color-coded by segment signal-to-noise. The minimum (33.3 ± 2.6 ms in this realization) lies on the pulse rise; background-dominated segments (red) trace the counting-noise ceiling, not the burst.

asymmetric-Gaussian peak model (Bernardini et al. 2015), and flux-randomization Monte Carlo errors (Peterson et al. 1998), taking the MC median with 16/84-percentile uncertainties, scanning the fit window around the peak so the window spread enters as a quoted systematic: $\tau = 0.725^{+0.070}_{-0.094} \pm 0.139_{\text{win}}$ s, soft lagging hard (positive lag in the standard convention; Norris et al.

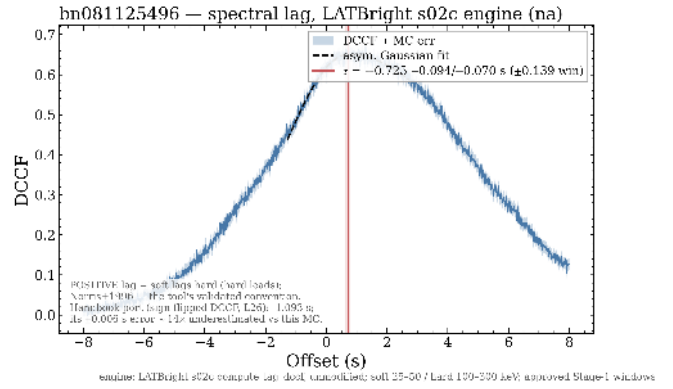


Figure 8. Discrete cross-correlation of the 25–50 and 100–300 keV light curves with the asymmetric-Gaussian peak model and the MC-median lag (positive = soft lags hard). For a single broad FRED the CCF width is the pulse width; the lag measures envelope evolution (§5.4).

1996; Ukwatta et al. 2010), with a 39σ CCF peak (Figure 8).¹

For a single broad pulse this number requires interpretation. The soft and hard light curves are near-copies of one envelope, so the CCF is a broad function whose width is set by the pulse duration; its peak offset measures the soft–hard shift of the envelope — the FRED’s hard-to-soft evolution — not a photon arrival delay. Consistently, $\tau/T_{90} \approx 0.085$, in line with lag-width scaling within the pulse paradigm (Norris et al. 1996; Ukwatta et al. 2012). This has a population consequence: in lag–variability correlations (Shenoy et al. 2013; Göktas et al. 2025), pulse width can enter both axes for broad-pulse bursts, and single-pulse samples with width-scaled windows (this campaign) are the clean way to separate the physics from the scaling.

6. TIME BINNING

We bin the source window with Bayesian blocks (Scargle et al. 2013) run through the Multi-Mission Maximum Likelihood framework (3ML; Vianello et al. 2015), followed by a significance-based merge requiring $S \geq 10$ per bin on the reference detector (cf. Yu et al. 2019, who apply $S \geq 20$ to detector-combined spectra; the single-detector floor is comparable after the $\sqrt{M}$ correction). This yields nine time bins spanning -1.70 to 10.65 s (Figure 9).

7. TIME-RESOLVED SPECTROSCOPY

7.1. Method

¹ An earlier in-house lag product was quarantined after failing sign-convention validation against the standard Norris definition; the defect and its resolution are recorded in the campaign ledger.

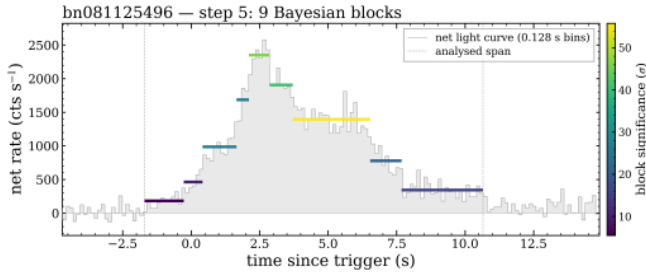


Figure 9. Time binning for GRB 081125496. The Bayesian-block decomposition of the reference-detector light curve (Scargle et al. 2013) is merged to the nine spectroscopy bins (shaded spans) by the $S \geq 10$ significance floor; block edges are drawn from the reference NaI.

For each of the nine bins and the time-integrated (T_{INT}) spectrum we fit 24 spectral models: the standard single-continuum forms (power law, cutoff power law, Band, smoothly broken power law, double smoothly broken power law (cf. Band et al. 1993; Ravasio et al. 2019)) and their combinations with blackbody, power-law, and cutoff power-law components. Fits minimize the Poisson–Gaussian likelihood (PGstat) in 3ML with inter-detector effective area corrections free within $[0.8, 1.2]$ for the non-reference detectors. Model selection uses the Akaike Information Criterion (AIC; Akaike 1974); we treat $\Delta\text{AIC} < 2$ as a statistical tie. As a representative goodness measure, the time-integrated winner has $\text{PGstat}/\text{dof} = 4226.4/342$.

7.2. Results

Table 1 lists the AIC winner and its parameters in every bin. Figure 10 and Figure 11 show the two flagship spectra — the time-integrated Band fit and the peak-bin (bin 4) cutoff power-law fit — in νF_ν representation with data unfolded following the XSPEC `eeufspec` construction (group-integrated ratio unfolding), 68% error bands from 3ML’s native error propagation, and count-space residuals. Figure 12 shows all 24 model fits for the time-integrated spectrum in a single AIC-ordered montage; equivalent montages for every bin are provided in Appendix A.

The spectral story is compact:

A cutoff power law suffices almost everywhere. CPL wins six of nine resolved bins (Table 1); in those bins every added component either rails to a boundary or reproduces the CPL shape. The margins to the runner-up are thin throughout ($\Delta\text{AIC} = 0.3\text{--}2.0$): this is a burst in which many shapes tie, and we report winners with that caveat.

A high-energy tail is demanded only on the rise and in the late tail. In bin 2 (the pulse rise) a pure cutoff leaves a coherent positive run across six of seven BGO groups at 0.3–10 MeV, and Band-family models

absorb it ($\beta = -2.08^{+0.13}_{-0.19}$). In bin 8 (the late tail) the tail demand returns decisively (Band $\beta = -2.18^{+0.17}_{-0.26}$, $E_p = 37^{+16}_{-7}$ keV). In between, the tail is absent or sub-threshold. The models tied at $\Delta\text{AIC} < 1$ in bin 8 disagree by an order of magnitude in 30 MeV flux: high-energy extrapolations from this burst are model-dominated and we do not quote them.

The time-integrated winner is an artifact of evolution. The smoothly broken power law plus power law wins T_{INT} (AIC = 4242.4, a statistical dead heat with three other composites) but wins *no* resolved bin, and its power-law component is invalid (railed) in eight of nine bins. Integrating a spectrum whose cutoff energy evolves by a factor $\gtrsim 5$ (Figure 13) manufactures an apparent second component. This is a concrete, single-burst demonstration of why time-integrated component claims require time-resolved confirmation.

No photospheric component. Across ~ 90 blackbody-added fits in all bins we find zero thermal evidence. The fitted blackbody population is bimodal: a $kT \simeq 1.1\text{--}1.9$ keV cluster whose Wien peak ($3.92kT \lesssim 7.5$ keV) lies *below* the 8 keV fitted boundary — edge-constrained artifacts absorbing low-energy calibration structure, not photospheres — and a $kT \simeq 10\text{--}40$ keV cluster with normalizations consistent with zero. The bin-2 AIC winner (Band+BB, $kT = 1.55^{+0.51}_{-0.37}$ keV) is itself the cautionary case: an edge-constrained feature at the top of a ranking table. We flag such solutions by the criterion $3.92kT$ below the fitted band and exclude them from any thermal census.

Inter-detector calibration. The n_b effective-area factor rests on its 0.8 lower bound in five of nine bins (e.g., 0.8134 ± 0.034 in bin 4); b_1 reaches 1.2 in two. The bounds are held fixed at $[0.8, 1.2]$ as a calibration prior. Low-energy features in this burst inherit that caveat, which is one reason the edge-constrained blackbodies above are not read as physical.

7.3. Parameter Evolution

For every model that wins at least one bin we track its parameters across all bins (Figures 13–16). The CPL panels carry the burst’s spectral history: the cutoff energy declines from $\simeq 320$ keV on the rise to $\simeq 59$ keV in the late decay — hard-to-soft evolution across a factor ~ 5.5 — while the low-energy index stays in a narrow range around $\Gamma \simeq -0.3$, well inside the synchrotron fast-cooling death line and consistent with marginally fast cooling at the peak bins. The Band panels show β railing to -5 (a CPL clone) everywhere except where the tail is real (bins 2, 5, 8). The SBPL+PL panel makes

Table 1. AIC-selected best model per time bin (statistical ties at $\Delta\text{AIC} < 2$ are noted in the text). Errors are MINOS asymmetric.

Bin	Interval (s)	Model	AIC	$\Delta\text{AIC}_{2\text{nd}}$	Parameters
T _{INT}	[−1.70, 10.65]	SBPL+PL	4242.4	0.8	$\alpha = -0.31^{+0.38}_{-0.35}$, $E_b = 146^{+58}_{-38}$, $\beta = -3.35^{+0.39}_{-0.54}$, PL: $-1.63^{+0.63}_{-0.15}$
0	[−1.70, −0.27]	CPL	1966.9	2.0	$\Gamma = -0.23^{+0.61}_{-0.42}$, $E_c = 315^{+260}_{-130}$
1	[−0.27, 0.42]	CPL	1207.0	0.8	$\Gamma = -0.43^{+0.27}_{-0.22}$, $E_c = 321^{+170}_{-100}$
2	[0.42, 1.66]	Band+BB ^a	1868.4	1.4	$\alpha = +0.02^{+0.31}_{-0.26}$, $E_p = 202^{+39}_{-28}$, $\beta = -2.08^{+0.13}_{-0.19}$, $kT = 1.55^{+0.51}_{-0.37}$
3	[1.66, 2.11]	CPL	813.1 ^b	1.6	$\Gamma = -0.31^{+0.13}_{-0.12}$, $E_c = 167^{+30}_{-24}$
4	[2.11, 2.87]	CPL	1374.2	1.5	$\Gamma = -0.147^{+0.090}_{-0.085}$, $E_c = 134^{+13}_{-12}$
5	[2.87, 3.71]	Band	1482.2 ^b	0.6	$\alpha = +0.08^{+0.17}_{-0.15}$, $E_p = 155^{+13}_{-12}$, $\beta = -2.99^{+0.36}_{-0.65}$
6	[3.71, 6.54]	CPL	2717.9	1.9	$\Gamma = -0.305^{+0.074}_{-0.072}$, $E_c = 77.7^{+5.8}_{-5.1}$
7	[6.54, 7.68]	CPL	1675.6	0.8	$\Gamma = -0.48^{+0.24}_{-0.22}$, $E_c = 59^{+16}_{-12}$
8	[7.68, 10.65]	Band	2646.0	0.3	$\alpha = -0.10^{+1.4}_{-0.93}$, $E_p = 36.9^{+16}_{-6.9}$, $\beta = -2.18^{+0.17}_{-0.26}$

^a Edge-constrained blackbody: $3.92 kT = 6.1$ keV lies below the 8 keV fitted boundary; not photospheric (§7.2).

^b AIC values are per-bin and not comparable across bins (different data).

NOTE—Energies in keV. All values provisional pending campaign freeze. The complete 24-model $\times$ 10-bin parameter tables are available as machine-readable products.

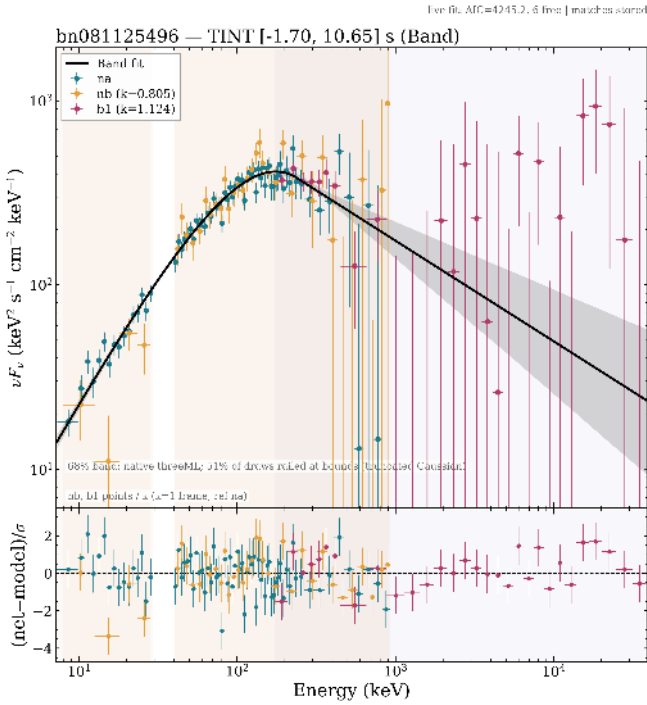


Figure 10. Time-integrated νF_ν spectrum with the Band fit. Data points are group-integrated ratio-unfolded (XSPEC `eeufspec` construction) at `rebin 5 5`; non-reference detectors are scaled by their fitted effective-area factors onto the $k = 1$ frame (factors in legend); the 68% band is 3ML’s native error propagation. Bottom: count-space residuals $(\text{net} - \text{model})/\sigma$.

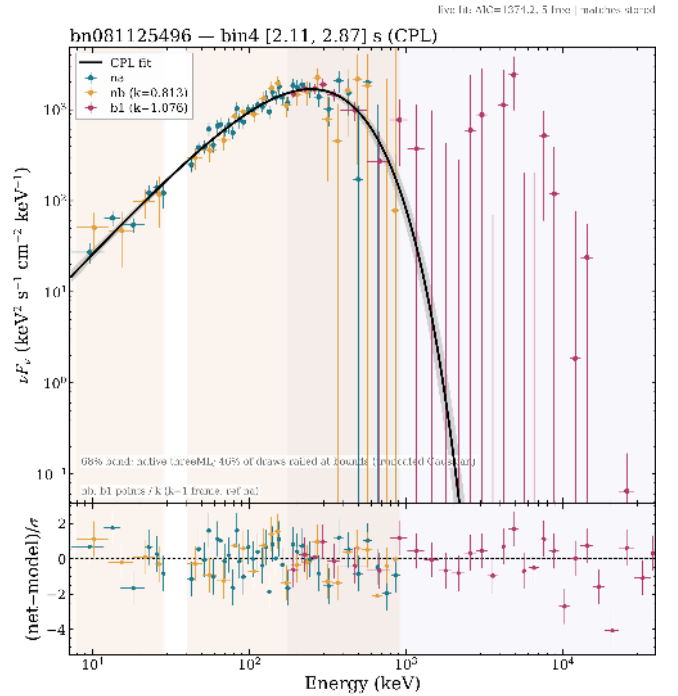


Figure 11. Same as Figure 10 for bin 4 with the winning cutoff power law.

§7.2’s point graphically: the model is valid in one bin only.

8. DISCUSSION

Assembled, the burst reads as one coherent object:

bn081125496 — TINT [-1.70, 10.65] s — 24 models, AIC-ordered — winner SBPLPL

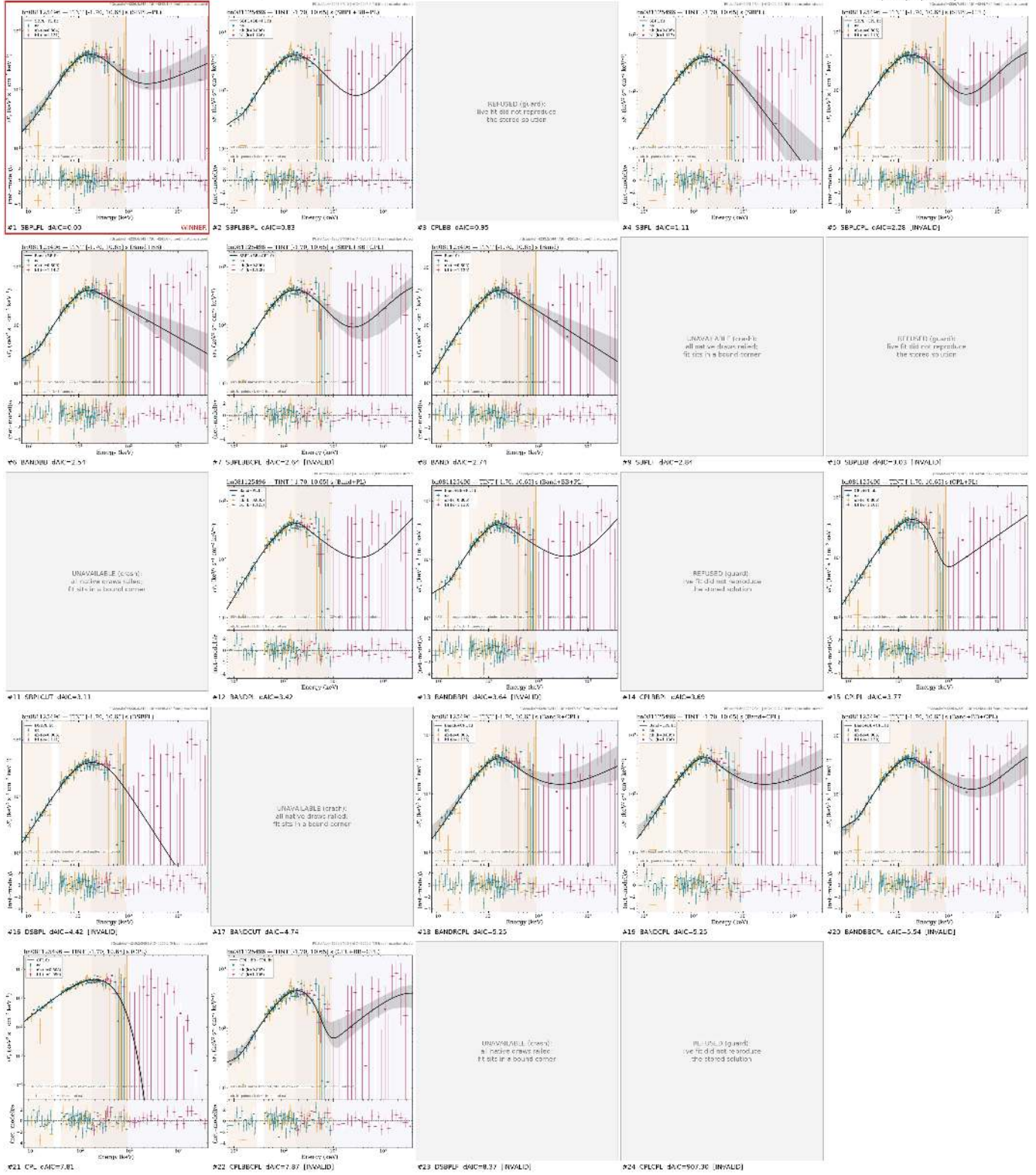


Figure 12. All 24 model fits to the time-integrated spectrum, AIC-ordered (winner top-left, red frame). Refused panels are labeled with the reason (live fit did not reproduce the stored solution, or degenerate error propagation); absence is never silent. Per-bin montages: Appendix A.

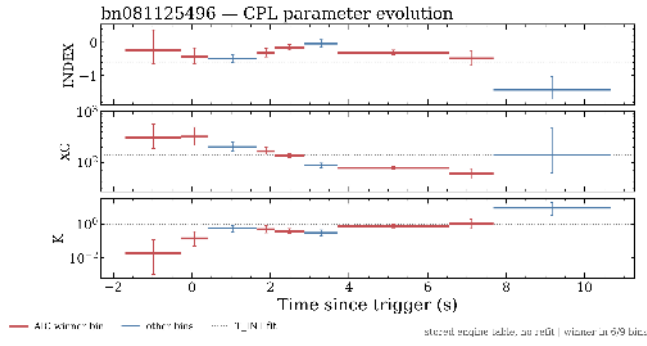


Figure 13. CPL parameter evolution across the nine bins. Top to bottom: low-energy index Γ , cutoff energy E_c , and normalization. Horizontal bars span each time bin; red bars mark bins where CPL is the AIC winner; the dotted line is the time-integrated value. E_c carries the burst’s hard-to-soft history while Γ stays near -0.3 .

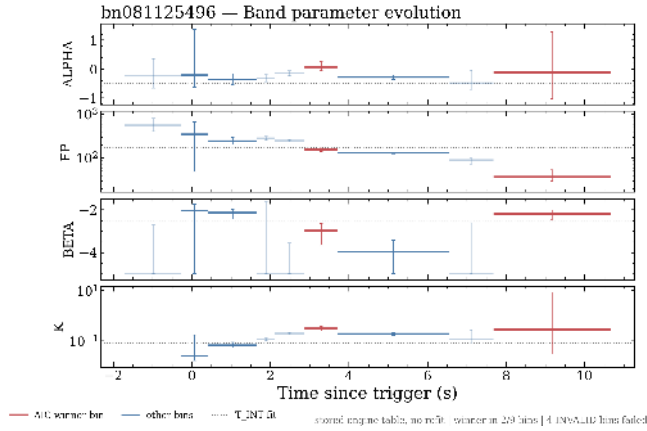


Figure 14. As Figure 13 for the Band function. Faded bars are fits failing validity gates (railed parameters).

One pulse, one continuum, evolving. A cutoff power law with $\Gamma \simeq -0.3$ and a decaying E_c describes seven of nine bins; the hard tail appears exactly where fast evolution concentrates photons above the cutoff (the rise) and where the decaying E_p drags the band window onto the tail (the late bins). The time-integrated composite winner is the integral of this evolution, not an extra emission component — a single-burst caution for integrated-spectrum component claims (cf. Yu et al. 2019).

No photosphere, and a reusable artifact class. The blackbody census is null, and its failure mode is instructive: every “significant” low- kT solution is pinned below the fitted band’s edge. A simple gate — flag any thermal component whose Wien peak falls below the fitted range — removes the entire class and will be applied across the campaign.

Timing quantities are estimator statements. The same data return MVTs of 34, 215, and 546 ms

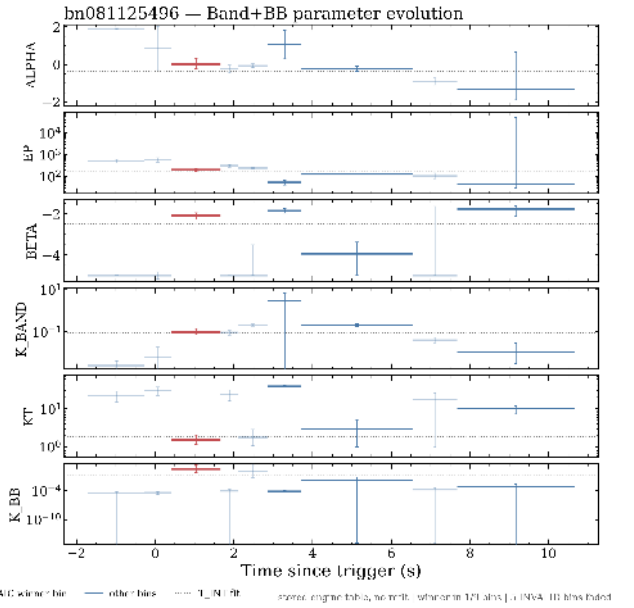


Figure 15. As Figure 13 for Band+BB. The kT panel shows the edge-constrained cluster near 1.5 keV (§7.2).

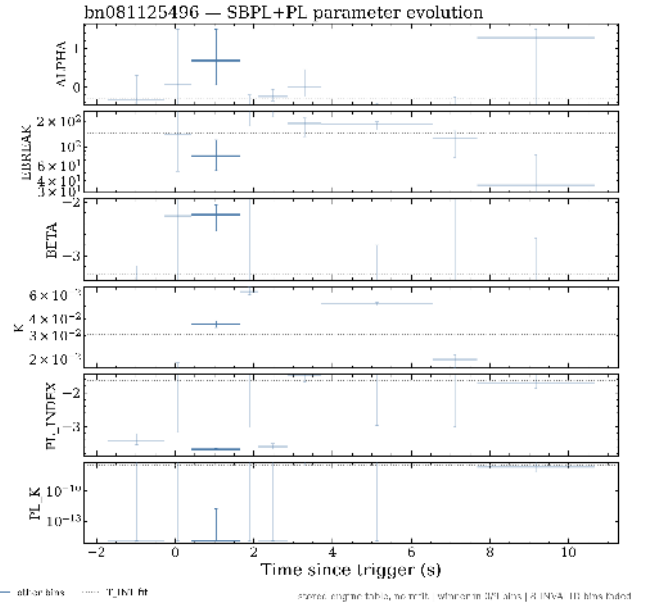


Figure 16. As Figure 13 for SBPL+PL, the time-integrated winner: valid in a single resolved bin.

under three published estimators, and a lag that is best read as 8% of the pulse width. None of these is wrong; each answers its own question. For population work — lag–luminosity (Ukwatta et al. 2012), lag–MVT curvature tests (Shenoy et al. 2013; Göktas et al. 2025), MVT compactness limits — the estimator, band, frame, and window must travel with every number, and single-pulse samples are where those choices can be controlled.

9. ANALYSIS PROVENANCE

Every figure in this paper passed a fresh-context verification gate (producer–verifier separation) before inclusion; every fitted value is bound to a machine-readable provenance sidecar (inputs, seeds, likelihood values, software hashes); live fits are hard-guarded to reproduce stored solutions to $|\Delta\text{AIC}| < 0.1$; and refused or degenerate fits are enumerated with reasons rather than omitted. The complete gate ledger and per-bin reviewer notes are archived with the campaign products. A companion paper describes the agentic pipeline and its verification architecture.

10. SUMMARY

We analyzed GRB 081125496 end-to-end with the single-pulse campaign pipeline:

1. $T_{90} = 8.50^{+0.15}_{-0.19}$ s (windowed lower limit; 5.3σ tail beyond the window), $T_{90} \propto E^{-0.14 \pm 0.02}$, FRED-like asymmetry $\varphi = 0.317 \pm 0.026$ (Kocevski profile best).
2. MVT = 33.9 ± 2.9 ms (windowed, on the pulse rise; $z = 2.5$ caveat), versus 215 ms (global CWT)

and 546 ms (global Haar) — estimator labels are part of the measurement.

3. Lag = $0.725^{+0.070}_{-0.094} \pm 0.139_{\text{win}}$ s (soft lags hard) $\approx 8.5\%$ of the pulse width: envelope evolution, with direct implications for population lag–variability correlations.
4. CPL suffices in 7/9 bins ($\Gamma \simeq -0.3$, E_c declining $321 \rightarrow 59$ keV); a hard tail ($\beta \simeq -2.1$) is demanded only on the pulse rise and in the late tail.
5. The time-integrated winner (SBPL+PL) wins no resolved bin — integrated composite components can be spectral evolution in disguise.
6. Zero photospheric evidence in ~ 90 blackbody-added fits; all low- kT solutions are edge-constrained artifacts, including one AIC winner.

This work used *Fermi*/GBM public data. Analysis performed with 3ML (Vianello et al. 2015). Draft produced within the single-pulse campaign’s staged, gate-reviewed pipeline; all numbers are provisional pending the campaign freeze.

APPENDIX

A. PER-BIN MODEL MONTAGES

Figures 17–25 show the complete 24-model montages for bins 0–8 (AIC-ordered; winner framed; refusals labeled with reasons).

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bn081125496 — bin0 [-1.70, -0.27] s — 24 models, AIC-ordered — winner CPL

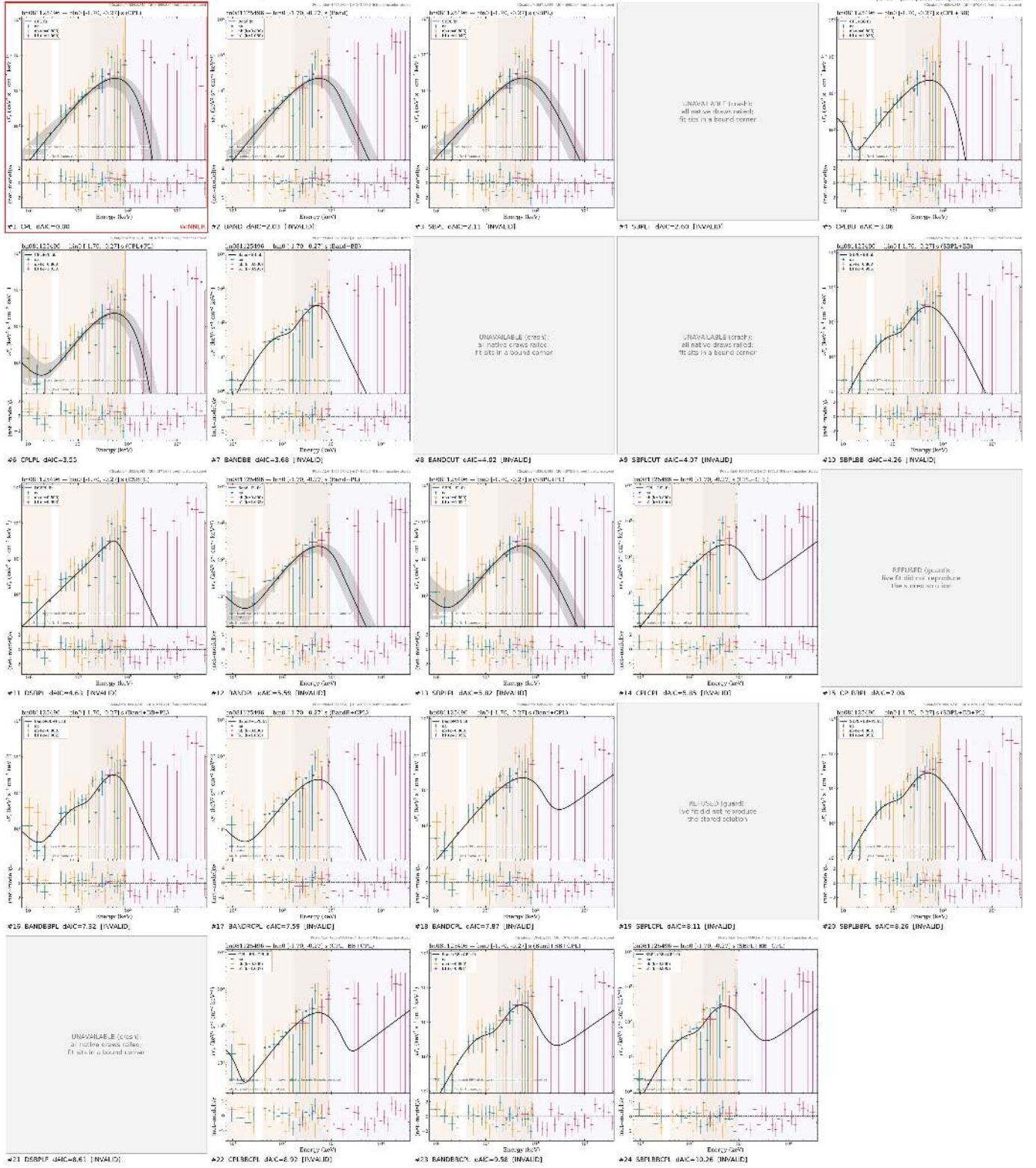


Figure 17. Bin 0 montage.

bn081125496 — bin1 [-0.27, 0.42] s — 24 models, AIC-ordered — winner CPL

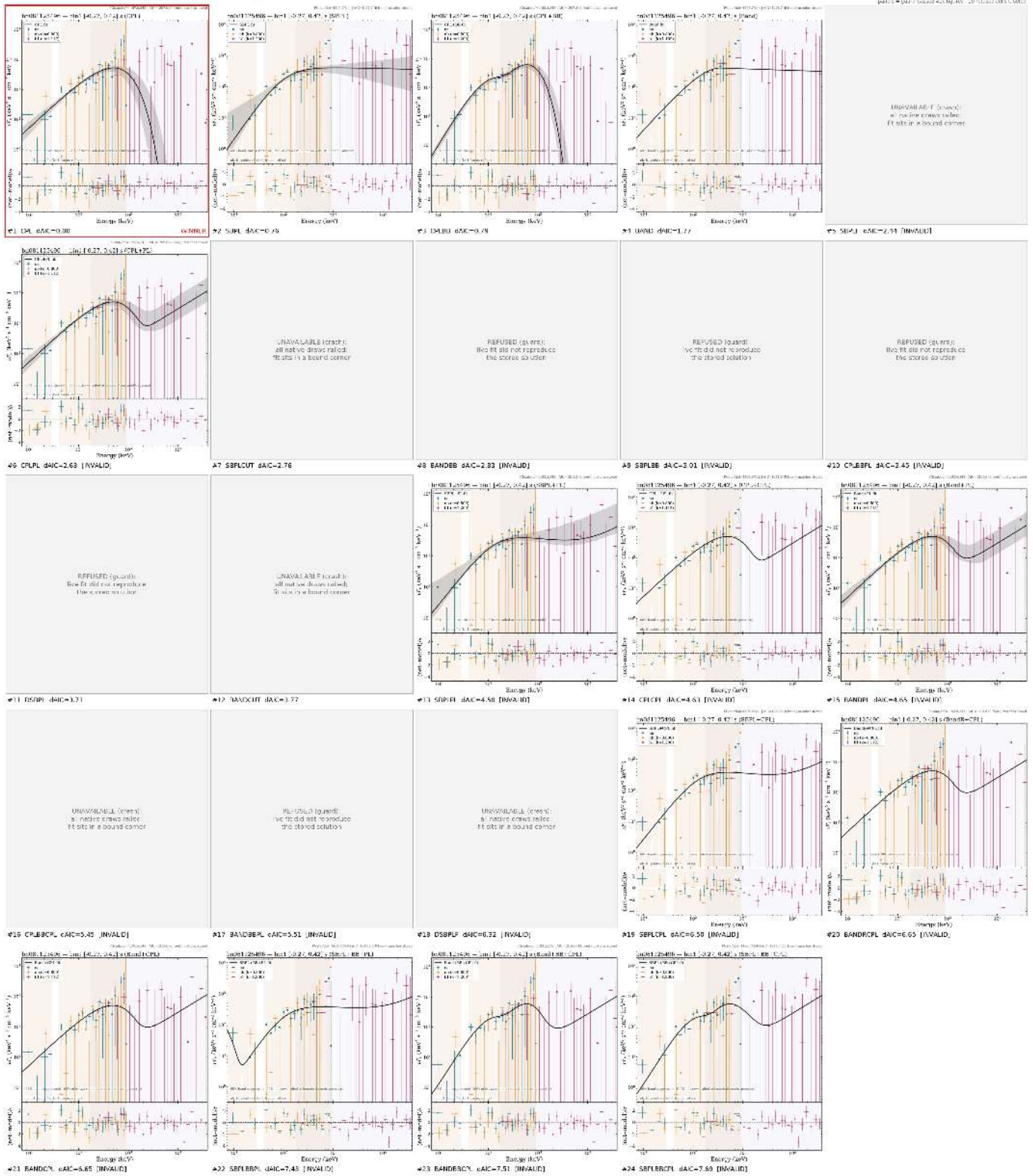


Figure 18. Bin 1 montage.

bn081125496 — bin2 [0.42, 1.66] s — 24 models, AIC-ordered — winner BANDDB

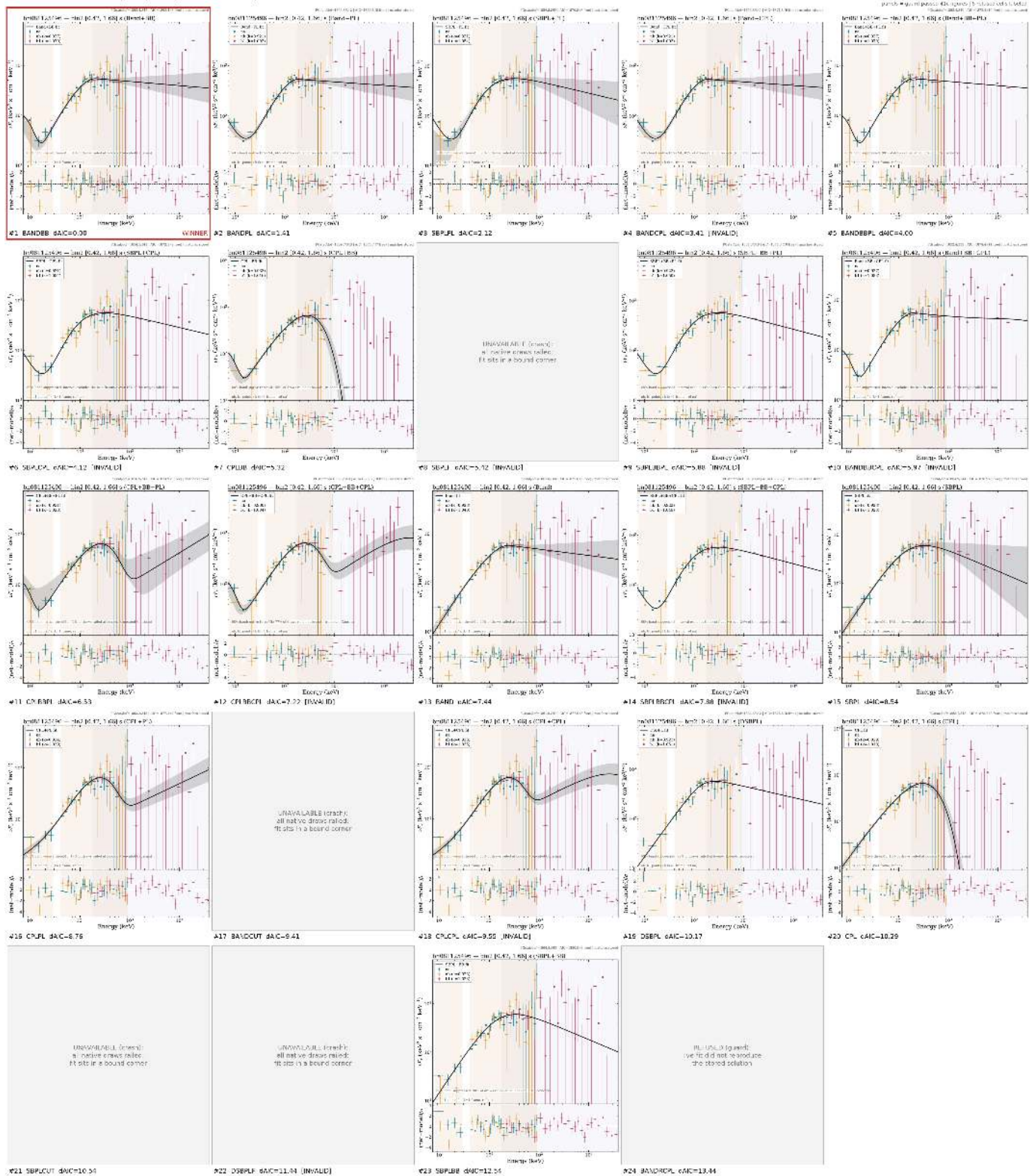


Figure 19. Bin 2 montage.

bn081125496 — bin3 [1.66, 2.11] s — 24 models, AIC-ordered — winner CPL

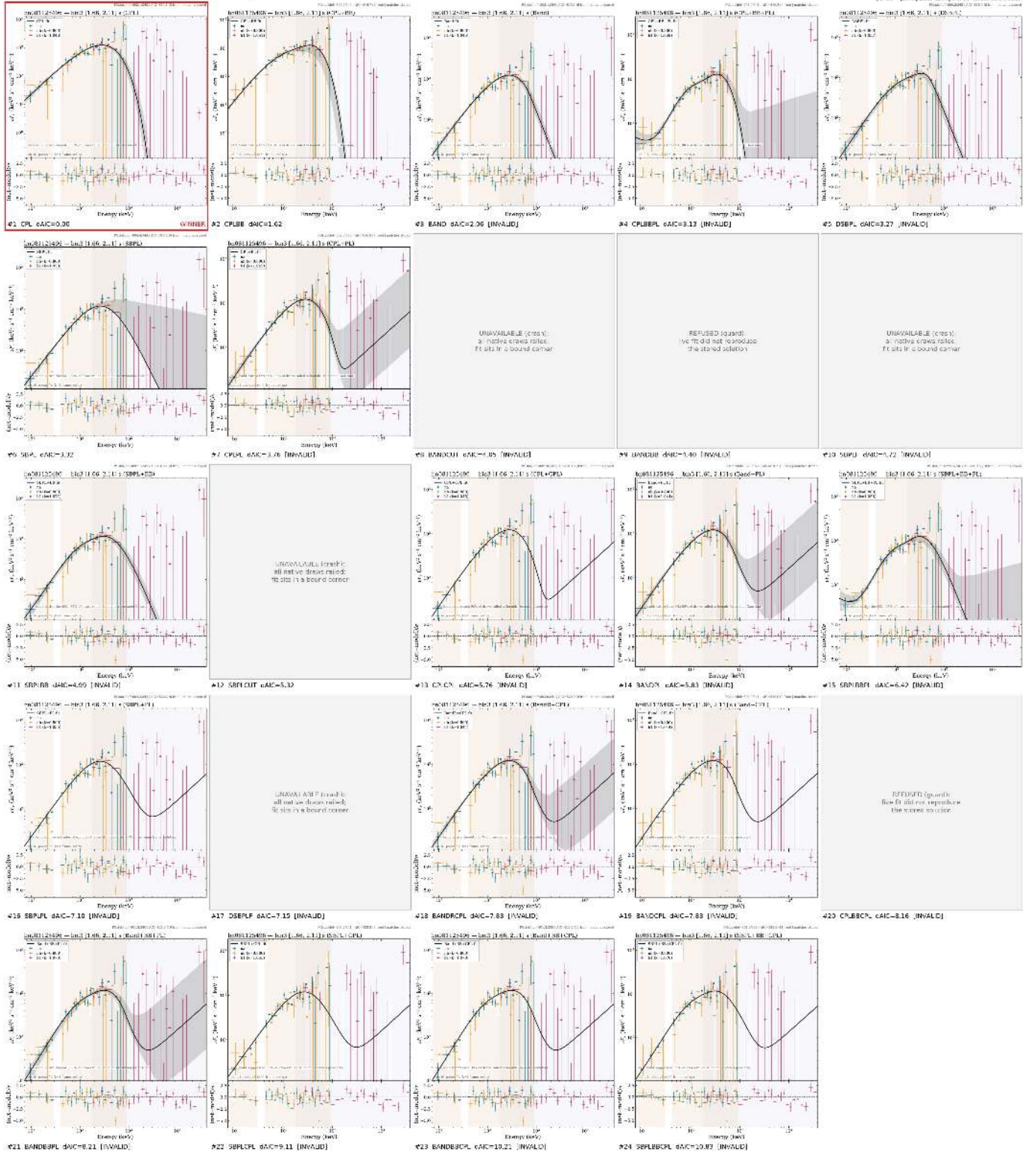


Figure 20. Bin 3 montage.

bn081125496 — bin4 [2.11, 2.87] s — 24 models, AIC-ordered — winner CPL

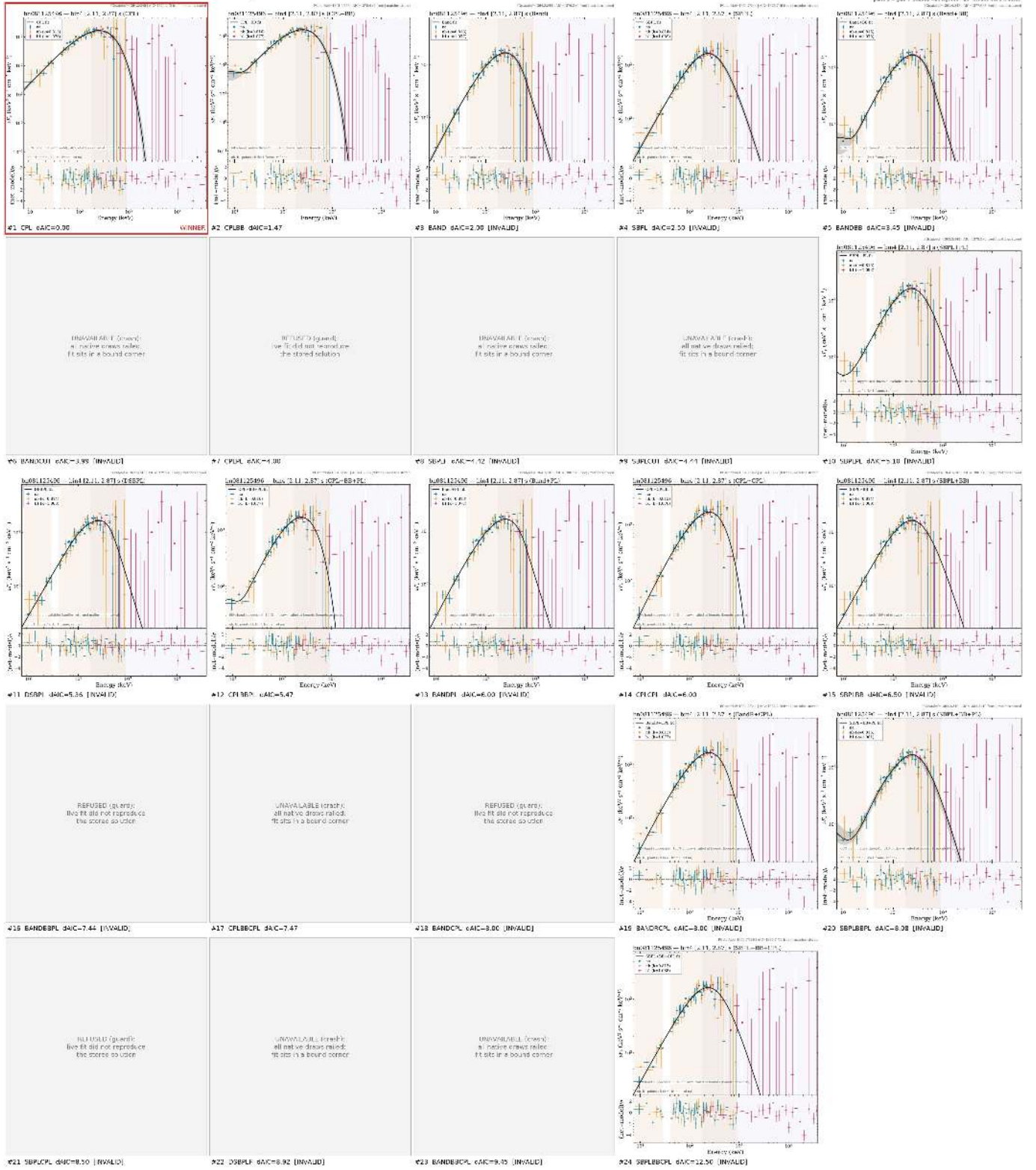


Figure 21. Bin 4 montage.

bn081125496 — bin5 [2.87, 3.71] s — 24 models, AIC-ordered — winner BAND

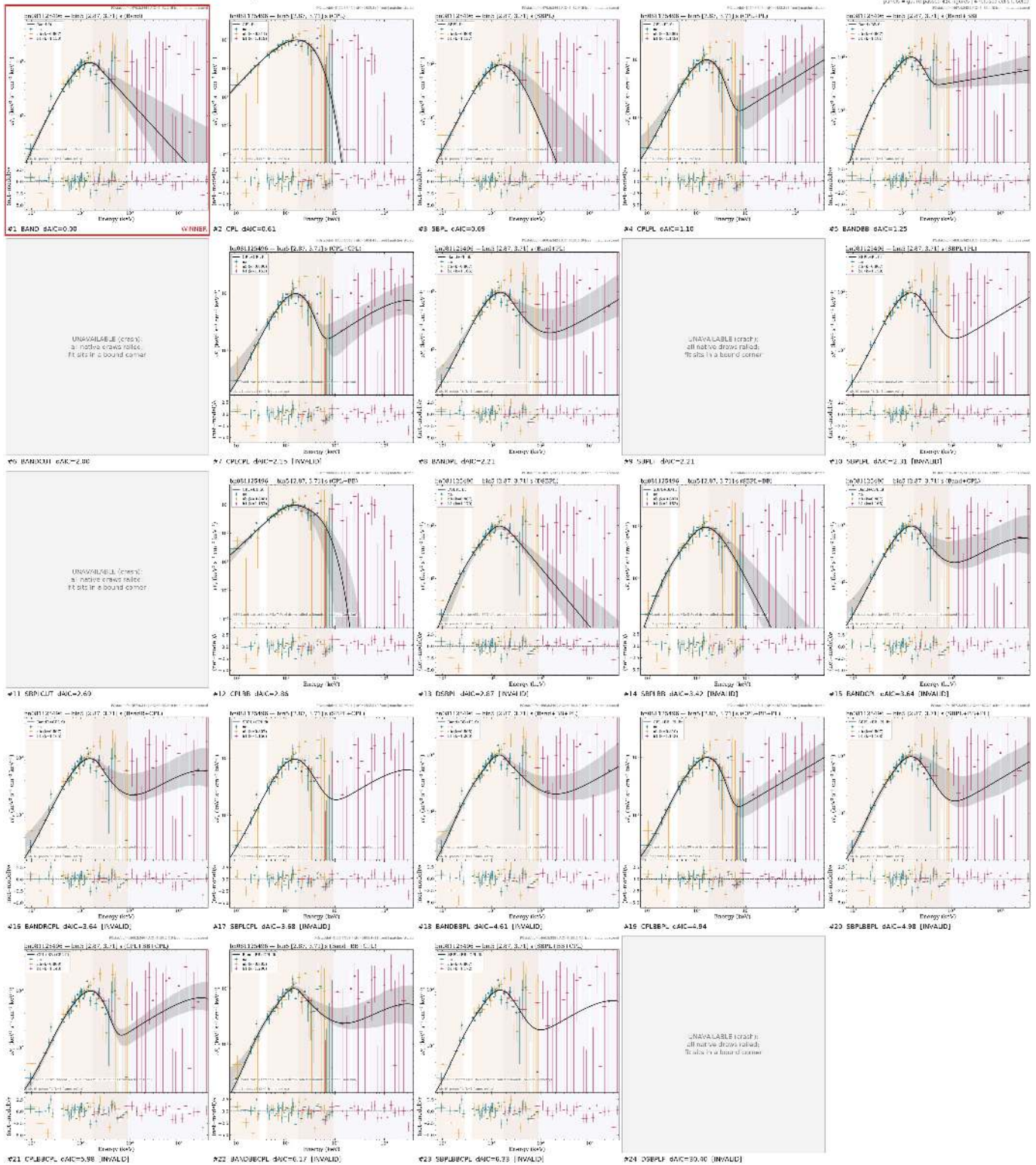


Figure 22. Bin 5 montage.

bn081125496 — bin6 [3.71, 6.54] s — 24 models, AIC-ordered — winner CPL

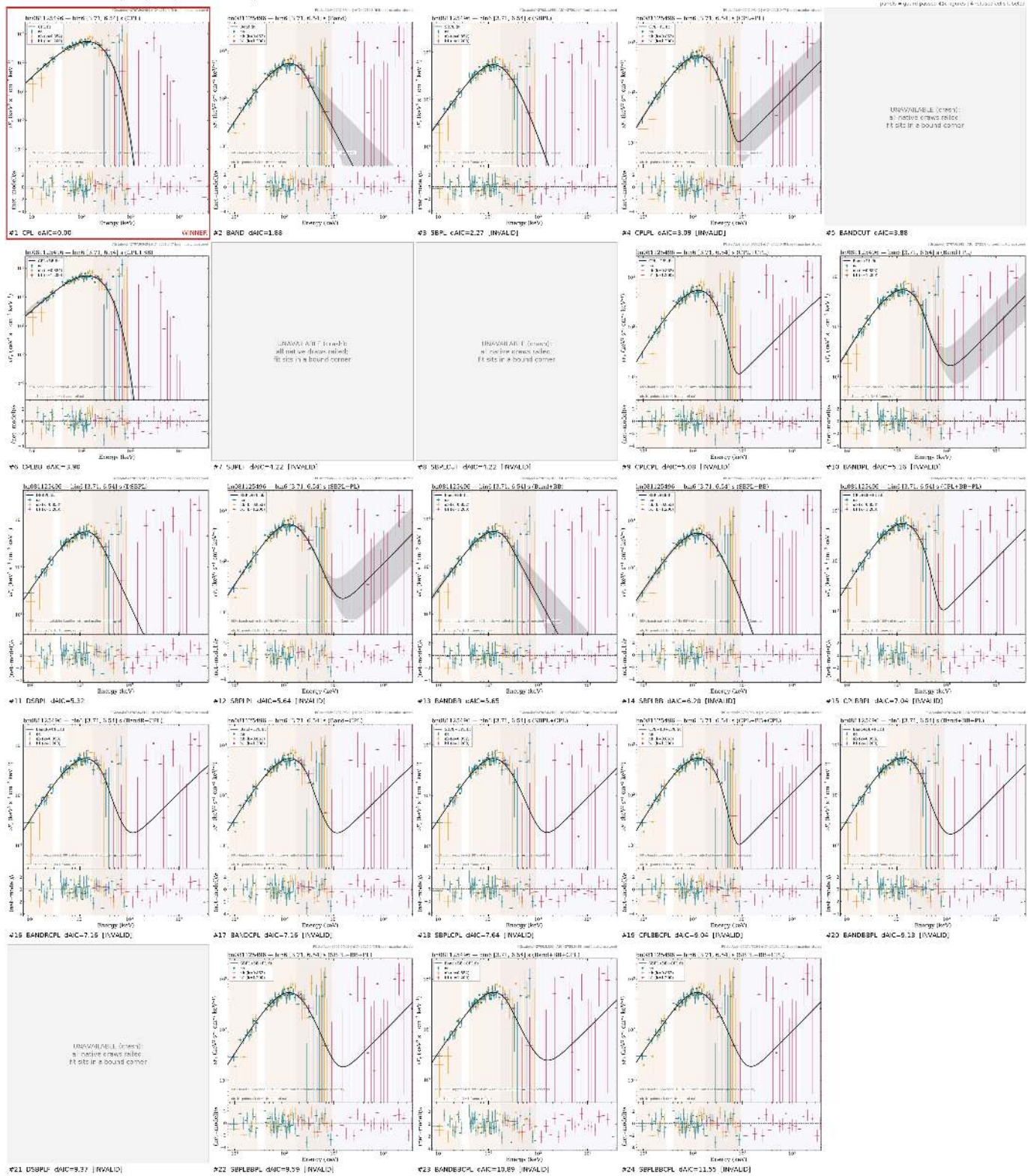


Figure 23. Bin 6 montage.

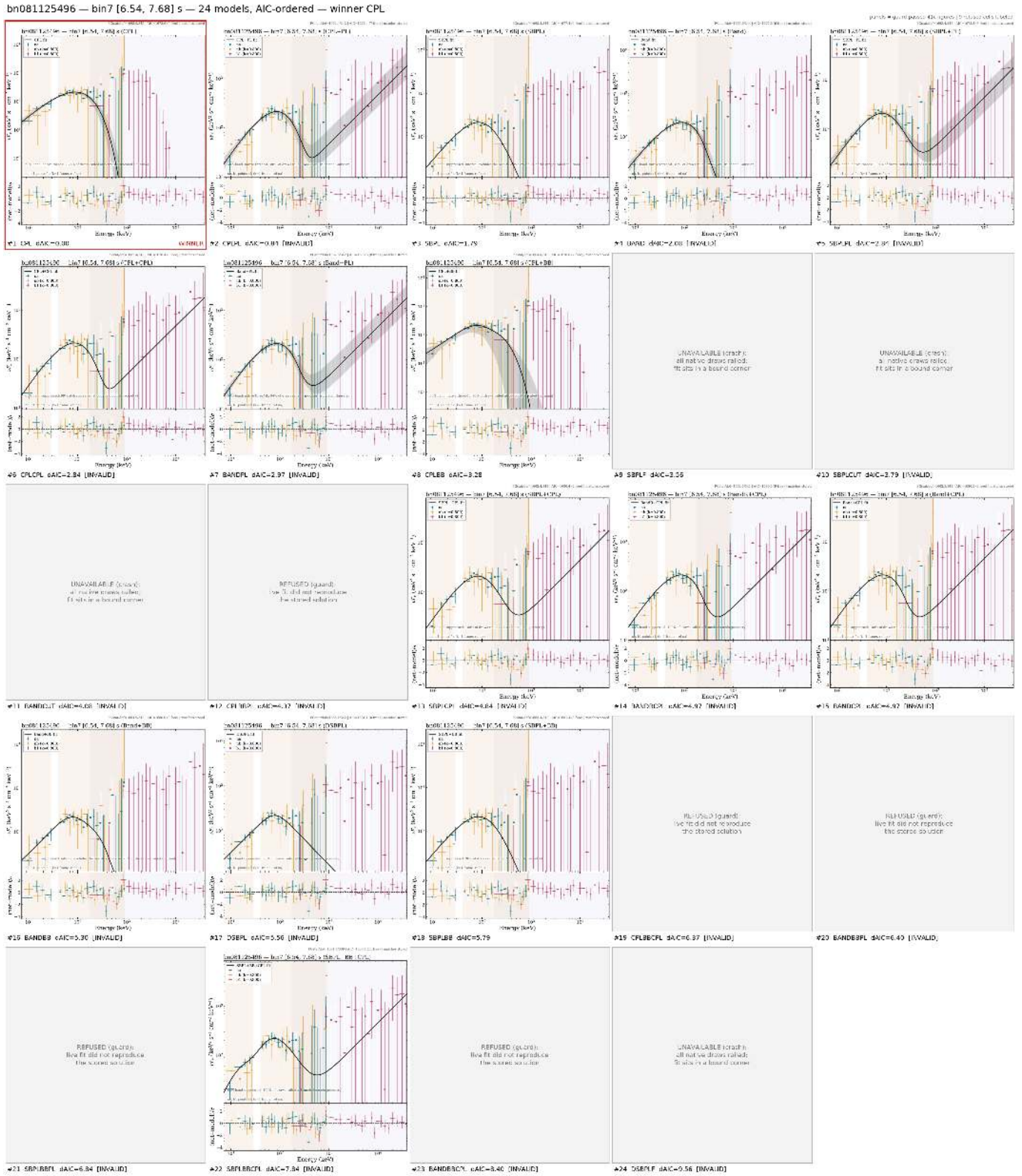


Figure 24. Bin 7 montage.

bn081125496 — bin8 [7.68, 10.65] s — 24 models, AIC-ordered — winner BAND



Figure 25. Bin 8 montage.

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